IT3-WavePixel v1.0

Three-Register Vacuum FSM, Neutron Strain Padding,
Chiral Boundary Orientation, Wilson-Holonomy Entanglement,
Spectral-Register Proofs for Selected Standard-Model Words,
Pb $\to$ Au Transmutation,
and the Unified Binary Register of Chemistry
and Particle Physics

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Abstract

We extend the IT3 wave-pixel architecture to a three-register finite-state model

$$(Z, N, M),$$

where Z is the nuclear hardware address, N is the neutron volume/ECC padding coordinate, and $M \in \{0,1\}^{172}$ is the electronic software occupation mask. The residual atomic topological charge is

$$Q_{\text{top}}(Z, M) = Z - \text{wt}(M).$$

Nuclear transmutation edits Z, ionization and spectroscopy edit M, while neutron emission or capture edits N without changing chemical identity. This gives exact register-level accounts of electron capture, β^--decay, β^+-decay, α-decay, ultra-peripheral neutron knockout, photoionization, and the Auger effect.

The channel

$$\text{Pb}\ (Z = 82) \longrightarrow \text{Au}\ (Z = 79) + 3p$$

is used as a charge-level transmutation archetype. At the register level it is a hardware address edit

$$\texttt{0x52} \longrightarrow \texttt{0x4F},$$

with binary address friction

$$\text{TFI}_3(82) = \text{popcount}(82 \oplus 79) = \text{popcount}(29) = 4, \qquad 29 = \texttt{0x1D}.$$

If the electronic mask is momentarily unchanged, the daughter register has

$$Q_{\text{top}} = 79 - 82 = -3,$$

so neutralization requires three electronic data deletions. For stable ^{197}Au from ^{208}Pb, the mass register must also lose eight neutrons:

$$208 - 3p - 8n = 197.$$

We introduce two exact neutron-padding proxies:

$$\Sigma_{\mathrm{strain}}(Z,N)=\frac{Z^2}{Z+N},\qquad \eta_{\mathrm{asym}}(Z,N)=\frac{N-Z}{Z+N}.$$

For fixed Z, increasing N lowers the active-node strain proxy Σ_{strain} and raises the asymmetry buffer η_{asym}. This gives a finite-register reading of why heavy stable nuclei require $N>Z$. The interpretation is explicitly [Level B]/[Level C], not a derivation of the semi-empirical mass formula or nuclear magic numbers.

We further formalize the lepton valency $v_\ell=2$ as a two-strand chiral register with orientation label $\epsilon_\ell=\pm1$. The weak charged current is read as routing through one oriented strand:

$$\Pi_\ell^{\mathrm{act}}=\frac{1-\epsilon_\ell}{2}.$$

Parity flips $\epsilon_\ell\mapsto-\epsilon_\ell$, hence

$$\Pi_\ell^{\mathrm{act}}\mapsto\frac{1+\epsilon_\ell}{2}\neq\Pi_\ell^{\mathrm{act}}.$$

This is an exact algebraic parity-breaking projector under the chiral-thread postulate, while the dynamical origin of the selected orientation remains [Level O].

Finally, compact torus holonomy on

$$T^3(1,\sqrt{2},\sqrt{3})/\mathbb{Z}_2$$

and Wilson-loop language are used to reinterpret entanglement as a shared global spectral thread rather than superluminal signalling. Bell-type nonseparability is not denied; it is given a finite-register geometric reading compatible with no-signalling.

The paper also strengthens the bridge to selected particle-physics register words. Using the IT3 spectral sector with valencies

$$v_\ell=2,\qquad v_q=3,\qquad v_{\mathrm{total}}=5,\qquad \dim I=v_\ell v_q=6,$$

we prove the exact register identities

$$F_{3\mathrm{loop}}=v_{\mathrm{total}}(\dim I)^2+(v_q^2-v_\ell^2)=5\cdot36+5=185,$$

and, from the Chern–Simons level $k_{\mathrm{CS}}=v_\ell=2$,

$$\Delta_e=\frac{k_{\mathrm{CS}}}{v_\ell}=1,\qquad \Delta_q=\frac{k_{\mathrm{CS}}}{v_{\mathrm{total}}}=\frac{2}{5},$$

$$A_e=\frac{4\Delta_e}{4+\Delta_e^2}=\frac{4}{5},\qquad A_q=\frac{4\Delta_q}{4+\Delta_q^2}=\frac{5}{13},$$

hence

$$A_{\mathrm{FB}}=\frac{3}{4}A_eA_q=\frac{3}{13}.$$

In hexadecimal register form,

$$(3,13)=\big(\texttt{0x0003},\texttt{0x000D}\big).$$

The fine-structure register satisfies the exact integer identities

$$137=\texttt{0x89}=\texttt{0xF0}-\texttt{0x67},$$

$$137=13^2-32,\qquad 137=136\vee1.$$

Under the IT3 spectral axioms one further obtains

$$\alpha_{\mathrm{geom}}^{-1}=\Lambda_1^2+\frac{N_{\mathrm{twist}}}{v_q}+\frac{\Lambda_3}{v_\ell v_q}+\frac{|\mathrm{Wr}|_{\mathrm{dyn}}}{v_\ell v_q v_{\mathrm{total}}}\approx137.037,$$

whose truncated integer part is the exact register word $137=\texttt{0x89}$.

The heavy-vector resonance is represented by the mass-bus word

$$M_{Z'} \leftrightarrow 1088 \text{ GeV} = \texttt{0x0440},$$

with the spectral-register structure

$$M_{Z'} = \eta(D'_\gamma)\, M_{\text{bulk}}\, E_{\text{egg}}^{-1}\, \det{}_F(I - \alpha_{\text{geom}} P_\gamma),$$

where, in the rank-one projection sector,

$$\eta(D'_\gamma) = v_\ell = 2, \qquad E_{\text{egg}}^{-1} = \sqrt{3} + \sqrt{2}, \qquad \det{}_F(I - \alpha_{\text{geom}} P_\gamma) = 1 - \alpha_{\text{geom}}.$$

For the doubly charmed baryon, the truncated linear/experimental register pair is

$$3619/3621 = \big(\texttt{0x0E23}, \texttt{0x0E25}\big),$$

with residual

$$3621 - 3619 = 2 = \texttt{0x0002}.$$

The confinement bound gives

$$\frac{S_{\text{macro}}}{S_{\text{micro}}} = \frac{72}{24} = 3 = \texttt{0x0003}.$$

The result is a finite-state architecture in which nuclear physics edits the address, neutrons edit the volume buffer, chemistry edits the data mask, spectroscopy reads the XOR-diff, parity violation is a chiral projector, entanglement is global holonomy, and selected Standard-Model observables appear as register words.

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1 Introduction: from two registers to a three-register FSM

The IT3-WavePixel v0.3 architecture separated two kinds of edits:

$$\text{hardware address edit: } Z, \qquad \text{software data edit: } M.$$

This already explained the lead-to-gold intuition:

$$\mathrm{Pb}\ (Z = 82) \to \mathrm{Au}\ (Z = 79) + 3p$$

as a binary address edit

$$\texttt{0x52} \to \texttt{0x4F}.$$

However, nuclear physics contains another logically distinct coordinate: the neutron number N. Neutrons change the mass number

$$A = Z + N$$

but do not change the nuclear charge Z. Therefore they do not change the chemical element. This motivates the three-register state of IT3-WavePixel v1.0:

$$\boxed{(Z, N, M)}$$

where

- Z is the nuclear hardware address;
- N is the neutron volume/ECC padding coordinate;
- $M \in \{0,1\}^{172}$ is the electronic software occupation mask.

The topological charge of the residual atomic register is

$$Q_{\mathrm{top}}(Z, M) = Z - \mathrm{wt}(M).$$

Neutron edits leave Q_{top} unchanged:

$$V_m :\ (Z, N, M) \longmapsto (Z, N - m, M), \qquad \Delta Q_{\mathrm{top}} = 0.$$

This single extension allows us to treat electron capture, beta decay, alpha decay, ultra-peripheral neutron knockout, photoionization, and the Auger effect in one register grammar.

Version v1.0 adds four further architectural layers:

1. a neutron strain-padding proxy explaining qualitatively why heavy nuclei need $N > Z$;
2. a chiral lepton projector giving an algebraic form of parity violation;
3. a Wilson-holonomy reading of entanglement on the compact vacuum torus;
4. explicit spectral-register proofs for selected Standard-Model words: $F_{3\mathrm{loop}} = 185$, $A_{\mathrm{FB}} = 3/13$, the fine-structure register, the Z' mass word, the Ξ_{cc} residual, and the confinement ratio.

Epistemic Guard 1.1. The integer and Hamming-weight identities are [Level A]. The interpretation of Z, N, M as hardware address, ECC padding, and software mask is [Level B], conditional on the IT3 wave-pixel postulates. The derivation of nuclear binding energies, weak-interaction dynamics, and Bell correlations remains [Level O]. The channel $\mathrm{Pb} \to \mathrm{Au} + 3p$ is a charge-level archetype; real fragmentation experiments may involve neutron evaporation, beta-delayed branches, multifragmentation, and detector-selected final states.

2 Epistemic protocol

Status	Meaning
[Level A]	Exact arithmetic, binary, or cited theorem.
[Level B]	Conditional architectural interpretation under IT3 postulates.
[Level C]	Structural resonance: suggestive exact coincidence, not a derivation.
[Level O]	Open dynamical target.

The methodological rule is:

> We do not fit elements to numbers. We allocate wave-layers, then read elements as masks.

Nor do we claim that the register model derives the full Standard Model Lagrangian, the nuclear shell model, or quantum information theory. It gives a finite-state grammar in which selected phenomena can be represented exactly.

3 The 172-layer spectral pixel

3.1 Layer capacities

Let the floor index be $n = 0, 1, \ldots, 11$. Define

$$\kappa(n) = \min(n, 11 - n),$$

and

$$q_n = \left\lfloor \frac{\kappa(n) + 3}{2} \right\rfloor.$$

The binary holographic capacity law is

$$C(n) = 2q_n^2.$$

Explicitly,

$$(C_0, \ldots, C_{11}) = (2, 8, 8, 18, 18, 32, 32, 18, 18, 8, 8, 2).$$

Theorem 3.1 (Total pixel dimension [Level A])**.**

$$\sum_{n=0}^{11} C(n) = 172.$$

Moreover,

$$172 = 43 \ll 2.$$

Proof. Direct summation gives

$$2 + 8 + 8 + 18 + 18 + 32 + 32 + 18 + 18 + 8 + 8 + 2 = 172.$$

Since $43_2 = \texttt{101011}_2$,

$$43 \ll 2 = 172.$$

□

The cumulative closures are

$$S_N = \sum_{n=0}^{N} C(n),$$

hence

$$(S_0, \ldots, S_{11}) = (2, 10, 18, 36, 54, 86, 118, 136, 154, 162, 170, 172).$$

Denote the closure set by

$$\mathcal{S} = \{2, 10, 18, 36, 54, 86, 118, 136, 154, 162, 170, 172\}.$$

3.2 Elements as occupation masks

Choose a canonical order of the 172 wave modes. For $0 \leq Z \leq 172$, define the neutral elemental mask

$$M_Z \in \{0,1\}^{172}$$

as the prefix mask with the first Z modes occupied:

$$M_Z = (\underbrace{1, \ldots, 1}_{Z}, \underbrace{0, \ldots, 0}_{172-Z}).$$

Theorem 3.2 (Injectivity of the elemental map [Level A]given the order)**.** *If the canonical order is fixed, then*

$$Z \neq Z' \Longrightarrow M_Z \neq M_{Z'}.$$

Proof. The Hamming weights differ:

$$\mathrm{wt}(M_Z) = Z.$$

□

Thus an element is a wave-overlay: a distinct binary image inside the 172-layer spectral pixel.

Epistemic Guard 3.1. The prefix order used here is an audit order. The physical canonical order should be derived from a spectral action. That derivation is [Level O].

4 Three-register state space

Definition 4.1 (Atomic register state [Level B])**.** An atomic register state is a triple

$$(Z, N, M),$$

where

$$0 \leq Z \leq 172, \qquad N \in \mathbb{N}, \qquad M \in \{0,1\}^{172}.$$

Here

- Z is the nuclear hardware address;
- N is the neutron volume/ECC padding coordinate;

- M is the electronic software occupation mask.

The mass address is

$$A_{\text{mass}} = Z + N.$$

The topological charge of the residual atomic register is

$$Q_{\text{top}}(Z, M) = Z - \text{wt}(M).$$

Theorem 4.1 (Edit bookkeeping [Level A])**.** *For any process changing the nuclear address by ΔZ and the bound electronic weight by $\Delta\,\text{wt}(M)$, the residual topological charge changes by*

$$\Delta Q_{\text{top}} = \Delta Z - \Delta\,\text{wt}(M).$$

Proof. Immediate:

$$Q_{\text{top}}{}' - Q_{\text{top}} = (Z + \Delta Z) - (\text{wt}(M) + \Delta\,\text{wt}(M)) - (Z - \text{wt}(M)) = \Delta Z - \Delta\,\text{wt}(M).$$

□

Remark 4.1 (Total electric charge). The quantity Q_{top} is the charge of the residual atomic register. When emitted particles are included, total electric charge is conserved in the usual way. For example, in β^--decay the residual atom may have $Q_{\text{top}} = +1$, while the emitted electron carries charge -1.

5 Synchronized hardware/software edits

The three-register model gives exact arithmetic for several standard nuclear and atomic processes.

Process	ΔZ	$\Delta\,\text{wt}(M)$	ΔQ_{top}	IT3 reading
Electron capture $p + e^- \to n + \nu_e$	-1	-1	0	synchronized HW/SW edit; daughter remains neutral up to inner-shell vacancy
β^--decay $n \to p + e^- + \bar{\nu}_e$	$+1$	0	$+1$	hardware address rises; emitted electron is not yet bound; residual ion is positive
β^+-decay $p \to n + e^+ + \nu_e$	-1	0	-1	hardware address falls; residual atom initially has one excess electron
α-decay	-2	0	-2	hardware address falls by two; residual atom initially negative; emitted alpha carries $+2$
Proton knockout $3p$ emission	-3	0	-3	three-node hardware destruction; neutralization requires three electronic deletions
UPC neutron knockout $\gamma + {}^{208}\text{Pb} \to n + {}^{207}\text{Pb}$	0	0	0	neutron ECC padding edit; chemical identity unchanged
Photoionization	0	-1	$+1$	one-bit software deletion
Electron attachment	0	$+1$	-1	one-bit software addition
Auger double ionization	0	-2	$+2$	two software deletions at fixed hardware address

5.1 Electron capture as a synchronized edit

For electron capture,

$$p + e^- \to n + \nu_e.$$

At the register level,

$$Z \to Z - 1, \qquad \mathrm{wt}(M) \to \mathrm{wt}(M) - 1.$$

Therefore

$$\Delta Q_{\mathrm{top}} = (-1) - (-1) = 0.$$

Thus the daughter atom is born neutral at the level of total electronic weight, although it has an inner-shell vacancy. That vacancy is then repaired by software hops and possibly Auger emission.

Example 5.1. For a neutral lead-like start,

$$(Z, \mathrm{wt}(M)) = (82, 82), \qquad Q_{\mathrm{top}} = 0.$$

After electron capture,

$$(Z', \mathrm{wt}(M')) = (81, 81), \qquad {Q_{\mathrm{top}}}' = 0.$$

5.2 Beta-minus decay as hardware-first edit

For β^--decay,

$$n \to p + e^- + \bar{\nu}_e.$$

The nuclear address increases:

$$Z \to Z + 1.$$

The emitted electron is not initially part of the bound electronic mask, so

$$\mathrm{wt}(M) \to \mathrm{wt}(M).$$

Therefore

$$\Delta Q_{\mathrm{top}} = (+1) - 0 = +1.$$

The residual atom is a positive ion until it captures an electron from the environment.

Example 5.2. Starting from neutral lead-like register,

$$(82, 82), \qquad Q_{\mathrm{top}} = 0.$$

After β^--decay,

$$(83, 82), \qquad Q_{\mathrm{top}} = +1.$$

5.3 UPC neutron knockout as pure ECC edit

In ultra-peripheral heavy-ion collisions, a photon can knock out a neutron:

$$\gamma + {}^{208}\mathrm{Pb} \to n + {}^{207}\mathrm{Pb}.$$

Here

$$Z = 82$$

is unchanged, and the bound electronic mask is unchanged to leading order. Only

$$N : 126 \to 125$$

changes.
Therefore

$$\Delta Q_{\mathrm{top}} = 0.$$

The chemical element remains lead. This is the cleanest architectural demonstration that neutrons are not address bits.

5.4 Auger effect as double software edit

The Auger effect begins with core ionization:

$$\mathrm{wt}(M) \to \mathrm{wt}(M) - 1, \qquad Q_{\mathrm{top}} \to +1.$$

An outer electron hops into the core hole:

$$R_{ij} : M \to M \oplus e_i \oplus e_j.$$

This preserves weight:

$$\mathrm{wt}(M) \to \mathrm{wt}(M), \qquad Q_{\mathrm{top}} \to Q_{\mathrm{top}}.$$

The transition energy is transferred to another electron, which is ejected:

$$\mathrm{wt}(M) \to \mathrm{wt}(M) - 1.$$

Net result:

$$\Delta Z = 0, \qquad \Delta\,\mathrm{wt}(M) = -2, \qquad \Delta Q_{\mathrm{top}} = +2.$$

Thus the Auger effect is a two-bit software deletion at fixed hardware address.

6 Neutron as ECC padding and topological strain buffer

Definition 6.1 (Neutron edit [Level A]/[Level B])**.** For integer m, define the neutron volume edit

$$V_m : \ (Z, N, M) \longmapsto (Z, N - m, M).$$

It changes the mass address

$$A = Z + N$$

but leaves the topological charge invariant:

$$Q_{\text{top}}(Z, M) = Z - \text{wt}(M).$$

This gives a precise architectural distinction:

Protons change the address; neutrons change the volume buffer.

A useful charge-density proxy is

$$\chi(Z, N) = \frac{Z}{Z + N}.$$

For fixed Z, increasing N lowers χ. This mirrors the empirical fact that heavy stable nuclei require $N > Z$.

6.1 Two exact strain proxies

Definition 6.2 (Neutron strain proxies [Level A])**.** For $Z > 0$, $N \geq 0$, define

$$\Sigma_{\text{strain}}(Z, N) = \frac{Z^2}{Z + N},$$

and

$$\eta_{\text{asym}}(Z, N) = \frac{N - Z}{Z + N}.$$

The first proxy measures active-address density squared per mass node. The second measures neutron padding asymmetry.

Lemma 6.1 (Padding lowers active-node strain [Level A])**.** *For fixed Z, if $N_2 > N_1$, then*

$$\Sigma_{\text{strain}}(Z, N_2) < \Sigma_{\text{strain}}(Z, N_1).$$

Also,

$$\eta_{\text{asym}}(Z, N_2) > \eta_{\text{asym}}(Z, N_1).$$

Proof. Since Z^2 is fixed and $Z + N$ increases, the fraction $Z^2/(Z + N)$ decreases. Similarly,

$$\eta_{\text{asym}}(Z, N) = \frac{N - Z}{N + Z} = 1 - \frac{2Z}{N + Z},$$

which increases with N. □

Postulate 6.1 (Lattice-contact strain model [Level B]/[Level C])**.** In the IT3 vacuum lattice, active address nodes Z create local metric tension. A crude contact-density proxy scales like

$$\frac{Z^2}{A}.$$

Stable registers require this strain to remain below a topological rupture threshold. Neutrons, being neutral padding nodes, increase A without increasing Z, thereby lowering the strain proxy. Hence heavy elements require progressively more neutron padding.

Example 6.1. For ^{208}Pb,

$$Z = 82, \qquad N = 126, \qquad A = 208.$$

The strain proxy is

$$\Sigma_{\text{strain}}(82, 126) = \frac{82^2}{208} \approx 32.33.$$

If one artificially forced $N = Z = 82$, then

$$A = 164, \qquad \Sigma_{\text{strain}}(82, 82) = \frac{82^2}{164} = 41.00.$$

Thus neutron padding lowers the strain proxy by about 21%.

Nuclide	Z	N	A	N/Z	$\Sigma_{\text{strain}} = Z^2/A$
^{4}He	2	2	4	1.000	1.000
^{12}C	6	6	12	1.000	3.000
^{56}Fe	26	30	56	1.154	12.07
^{208}Pb	82	126	208	1.537	32.33

The table is not a derivation of nuclear binding energies. It is a finite-register diagnostic: as the active address Z grows, the neutral padding coordinate N must grow faster than Z to keep the local topological strain below rupture.

Epistemic Guard 6.1. This is not a derivation of the semi-empirical mass formula, nuclear shell closures, or magic numbers. The proxy $Z^2/(Z + N)$ is not the Coulomb energy and not the liquid-drop symmetry term. The identification of neutrons with ECC padding and strain buffers is [Level B]/[Level C]. Magic numbers such as $N = 126$ may be read as resonance buffers, but their full dynamical explanation remains ordinary nuclear physics plus an open IT3 target.

7 Chiral lepton thread and parity violation

The IT3 spectral-geometric framework assigns topological valencies

$$v_\ell = 2, \qquad v_q = 3.$$

Equivalently, the finite algebra contains

$$\mathcal{A}_F = \mathbb{C} \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}),$$

where $M_2(\mathbb{C})$ acts on the lepton doublet bundle and $M_3(\mathbb{C})$ acts on the quark triplet bundle.

The lepton bundle has two strands. Architecturally, those two strands may be assigned an orientation label

$$\epsilon_\ell = \pm 1.$$

The weak charged current can then be read as routing only through one oriented strand:

$$\Pi_\ell^{\text{act}} = \frac{1 - \epsilon_\ell}{2}.$$

If the active strand is chosen as $\epsilon_\ell = -1$, then

$$\Pi_\ell^{\text{act}} = 1.$$

The opposite strand has

$$\frac{1 + \epsilon_\ell}{2} = 0.$$

A parity transformation flips

$$\epsilon_\ell \longmapsto -\epsilon_\ell.$$

Therefore

$$\Pi_\ell^{\mathrm{act}} = \frac{1-\epsilon_\ell}{2} \quad \longmapsto \quad \frac{1+\epsilon_\ell}{2}.$$

Unless $\epsilon_\ell = 0$, which is impossible for an orientation label, these projectors are distinct.

Theorem 7.1 (Algebraic parity non-invariance of the chiral projector [Level A]given the projector)**.** *For $\epsilon_\ell = \pm 1$,*

$$\frac{1-\epsilon_\ell}{2} \neq \frac{1+\epsilon_\ell}{2}.$$

Hence a Hamiltonian containing only Π_ℓ^{act} is not invariant under $\epsilon_\ell \mapsto -\epsilon_\ell$.

Proof. If the two expressions were equal, then

$$1-\epsilon_\ell = 1+\epsilon_\ell,$$

so

$$\epsilon_\ell = 0,$$

contradicting $\epsilon_\ell = \pm 1$. □

Postulate 7.1 (Boundary orientation of the lepton helix [Level B])**.** The two-strand lepton bundle is geometrically helical. In an oriented vacuum manifold with bulk quark braid and boundary lepton helix, the charged weak current is a boundary routing operator. It couples to one induced orientation of the lepton helix. Right-handed charged leptons exist as register spectators but are not routed through the same charged-current hardware channel.

This gives a geometric language for the observed fact that the Standard Model charged weak interaction couples to left-handed fermion doublets and right-handed antifermion doublets.

Epistemic Guard 7.1. The projector algebra is [Level A]once Π_ℓ^{act} is defined. The identification of Π_ℓ^{act} with the physical charged weak current is [Level B]. This scalar orientation projector is an IT3 register-level analogue of chiral routing. It is not the full γ_5 projector of the Dirac spinor bundle, although it mirrors the same logical structure: one oriented strand is active, the opposite strand is inert for the charged current. The deeper reason why the induced orientation is the one selected by nature remains [Level O].

8 Torus holonomy, Wilson loops, and entanglement as shared global thread

The IT3 vacuum fibre is compact:

$$\mathcal{M} \cong T^3(1, \sqrt{2}, \sqrt{3})/\mathbb{Z}_2.$$

Allowed discrete photon wavevectors satisfy

$$\vec{k} \cdot \lambda \in 2\pi\mathbb{Z} \qquad \forall \lambda \in \Lambda,$$

so modes are labelled by integer triples

$$\mathbf{n} = (n_1, n_2, n_3) \in \mathbb{Z}^3.$$

Their squared norms are

$$\frac{\|\vec{k}\|^2}{(2\pi)^2} = n_1^2 + \frac{n_2^2}{2} + \frac{n_3^2}{3}.$$

For a general closed cycle

$$\lambda = m_1 e_1 + m_2 e_2 + m_3 e_3, \qquad (m_1, m_2, m_3) \in \mathbb{Z}^3,$$

single-valuedness gives

$$\vec{k} \cdot \lambda = 2\pi(n_1 m_1 + n_2 m_2 + n_3 m_3).$$

Thus the holonomy coefficient is the integer pairing

$$\langle \mathbf{n}, \mathbf{m} \rangle = n_1 m_1 + n_2 m_2 + n_3 m_3.$$

On a compact torus, a mode is a global object. Two spatially separated local readers may access restrictions of the same global mode.

8.1 Wilson-loop language

Let A be the vacuum $U(1)$ connection and let C be a closed cycle on the torus. The Wilson loop is

$$W(C) = \mathrm{Tr}\ \mathcal{P} \exp\left(i \oint_C A \cdot dx \right).$$

For an abelian $U(1)$ sector this reduces to the holonomy phase

$$\exp\left(i \oint_C A \right).$$

On T^3, noncontractible loops are classified by homology classes

$$[C] \in H_1(T^3, \mathbb{Z}) \cong \mathbb{Z}^3.$$

Thus a global topological degree of freedom can be read locally through different address pointers without requiring a signal propagating between the pointers.

Postulate 8.1 (Entanglement as shared Wilson-holonomy thread [Level B])**.** An entangled pair is not a pair of independent local bits communicating faster than light. It is a nonseparable global register state, or equivalently a shared topological thread / Wilson-holonomy degree of freedom on the compact vacuum manifold, read through two local address pointers.

This does not deny Bell-type correlations. Rather, it says that the correlations are encoded in a global degree of freedom, not in a superluminal signal.

Proposition 8.1 (No-signalling compatibility [Level B], architectural)**.** *Local edits of one pointer label do not constitute a controllable faster-than-light message, because the accessible statistics of the other pointer are determined by the shared global state and the local measurement context, not by instantaneous remote writing.*

Proof sketch. In ordinary quantum theory, local operations on one subsystem cannot change the reduced density operator of a spacelike-separated subsystem. The IT3 register reading replaces the abstract tensor-product language by a global-mode / holonomy language, but preserves the no-signalling constraint. □

Epistemic Guard 8.1. The statement "entanglement does not exist" is false and must not be used. The correct IT3 statement is:

Entanglement is re-described as nonseparability of a global torus register state.

The Wilson-holonomy picture is a geometric reinterpretation of nonseparability. It does not by itself derive the full quantum joint probability distribution, Bell-inequality violations, or Tsirelson bounds. Those remain [Level O]dynamical targets. The architectural requirement is only that the picture preserve no-signalling and reproduce the operational statistics of quantum mechanics in the appropriate limit.

9 Hardware register: Pb → Au as address edit

9.1 Address-edit operator

Definition 9.1 (Hardware transmutation operator [Level B])**.** For integer k, define

$$H_k : \ Z \longmapsto Z - k.$$

At the neutral-mask level,

$$H_k(M_Z) = M_{Z-k}.$$

For lead to gold,

$$k = 3, \qquad Z = 82,$$

therefore

$$H_3(M_{82}) = M_{79}.$$

This is the formal statement:

$$\boxed{\mathrm{Pb} \to \mathrm{Au} + 3p \quad \Longleftrightarrow \quad M_{82} \to M_{79}.}$$

9.2 Binary audit of Pb → Au

The essential hexadecimal identities are:

$$82 = \texttt{0x52} = \texttt{0101 0010}_2,$$

$$79 = \texttt{0x4F} = \texttt{0100 1111}_2.$$

Subtraction of three protons is:

$$\texttt{0x52} - \texttt{0x03} = \texttt{0x4F}.$$

In low-nibble arithmetic:

$$\texttt{0x2} - \texttt{0x3}$$

requires a borrow from the high nibble:

$$\texttt{0x5} \to \texttt{0x4},$$

and the low nibble becomes

$$\texttt{0x12} - \texttt{0x03} = \texttt{0x0F}.$$

Therefore

$$\texttt{0x52} \to \texttt{0x4F}.$$

The XOR difference is

$$\texttt{0x52} \oplus \texttt{0x4F} = \texttt{0x1D}.$$

In decimal and binary:

$$82 \oplus 79 = 29 = \texttt{0x1D} = \texttt{0001 1101}_2.$$

The number of flipped address bits is

$$\text{popcount}(29) = 4.$$

Theorem 9.1 (Topological friction of the Pb $\to$ Au address edit [Level A])**.** *For the address edit* $82 \to 79$*,*

$$\text{TFI}_3(82) = \text{popcount}(82 \oplus 79) = 4.$$

Proof. Direct binary calculation:

$$82 \oplus 79 = 29 = \texttt{0x1D} = \texttt{0001 1101}_2,$$

which has four one-bits. □

This generalizes the IT3 binary friction law for α-decay:

$$\text{TFI}_2(Z) = \text{popcount}(Z \oplus (Z-2)).$$

Examples:

$$88 \to 86: \quad \text{TFI} = 3,$$

$$87 \to 85: \quad \text{TFI} = 1.$$

The present proton-knockout channel gives

$$82 \to 79: \quad \text{TFI} = 4.$$

9.3 Isotope and neutron-padding caveat

Removing only three protons from ^{208}Pb gives formally

$$Z = 79, \qquad A = 205,$$

i.e. a neutron-rich gold isotope, not stable ^{197}Au.
To obtain stable ^{197}Au from ^{208}Pb, one must also remove eight neutrons:

$$208 - 3p - 8n = 197.$$

Thus the full register edit is

$$(Z, N) = (82, 126) \longrightarrow (79, 118).$$

The hardware address changes by -3, and the neutron ECC padding changes by -8.

10 Layer location: Pb and Au live near the equatorial closure

The cumulative closure before floor 5 is

$$S_4 = 54,$$

and the equatorial closure is

$$S_5 = 86.$$

Both lead and gold lie in floor 5:

$$54 < 79 < 82 \leq 86.$$

Their offsets inside floor 5 are:

$$\Delta_{\mathrm{Au}} = 79 - 54 = 25,$$

$$\Delta_{\mathrm{Pb}} = 82 - 54 = 28.$$

Thus the local equatorial sequence is:

Element	Z	Hex	Binary	Floor	Offset in floor 5
Au	79	`0x4F`	`0100 1111`	5	25
Hg	80	`0x50`	`0101 0000`	5	26
Tl	81	`0x51`	`0101 0001`	5	27
Pb	82	`0x52`	`0101 0010`	5	28
Bi	83	`0x53`	`0101 0011`	5	29
Po	84	`0x54`	`0101 0100`	5	30
At	85	`0x55`	`0101 0101`	5	31
Rn	86	`0x56`	`0101 0110`	5	32

The operation

$$\mathrm{Pb} \to \mathrm{Au}$$

does not jump across a principal shell closure. It moves backward inside the same equatorial floor, from offset 28 to offset 25. In wave-pixel language, it destroys three local topological nodes near the equatorial boundary.

The central binary edge is still nearby:

$$86 = \texttt{0x56}, \qquad 87 = \texttt{0x57},$$

$$\texttt{0x56} \oplus \texttt{0x57} = \texttt{0x01},$$

$$\texttt{0x56} + \texttt{0x57} = \texttt{0xAD} = 173 = 172 + 1.$$

Thus the lead-gold region sits in the same equatorial neighbourhood that controls the classical closure $Z = 86$ and the absolute pixel closure $Z = 172$.

11 Software register: ions, electrons, and topological charge

11.1 Data-edit operators

Let $e_i \in \{0,1\}^{172}$ be the basis vector with a single one in position i.

Definition 11.1 (Single-bit data edit [Level A])**.** For a bit position i, define

$$D_i(M) = M \oplus e_i.$$

If bit i was occupied, D_i removes one electron:

$$\mathrm{wt}(D_i M) = \mathrm{wt}(M) - 1, \qquad Q_{\mathrm{top}} \to Q_{\mathrm{top}} + 1.$$

If bit i was empty, D_i adds one electron:

$$\mathrm{wt}(D_i M) = \mathrm{wt}(M) + 1, \qquad Q_{\mathrm{top}} \to Q_{\mathrm{top}} - 1.$$

Proposition 11.1 (Ionization as one-bit software edit [Level A]/[Level B])**.** *A one-electron ionization or attachment changes the electronic mask in exactly one bit and changes Q_{top} by ± 1.*

Proof. By definition, XOR with a basis vector flips exactly one bit. The Hamming weight changes by ± 1, hence $Q_{\mathrm{top}} = Z - \mathrm{wt}(M)$ changes by the opposite sign. □

11.2 Redox as weight-preserving paired data edits

Consider two atomic registers (Z_A, M_A) and (Z_B, M_B). An electron transfer from A to B changes

$$\mathrm{wt}(M_A) \to \mathrm{wt}(M_A) - 1, \qquad \mathrm{wt}(M_B) \to \mathrm{wt}(M_B) + 1.$$

Therefore the total electronic weight is conserved:

$$\mathrm{wt}(M_A) + \mathrm{wt}(M_B) = \mathrm{wt}(M_A') + \mathrm{wt}(M_B').$$

The topological charges change as

$$Q_{\mathrm{top}_A} \to Q_{\mathrm{top}_A} + 1, \qquad Q_{\mathrm{top}_B} \to Q_{\mathrm{top}_B} - 1.$$

Hence the total topological charge is conserved:

$$Q_{\mathrm{top}_A} + Q_{\mathrm{top}_B} = Q_{\mathrm{top}_A}' + Q_{\mathrm{top}_B}'.$$

Theorem 11.1 (Redox charge conservation [Level A])**.** *For any electron-transfer process with fixed nuclear addresses,*

$$\sum_i Q_{\mathrm{top}_i}$$

is conserved.

Proof. Each transferred electron decreases one mask weight by one and increases another by one. Thus the sum of weights is unchanged, while the sum of addresses is unchanged. Therefore the sum of $Z_i - \mathrm{wt}(M_i)$ is unchanged. □

12 Spectral diff: photons and internal transitions as mask patches

12.1 One-electron hop operator

Definition 12.1 (Hop operator [Level A])**.** If bit i is occupied and bit j is empty, define the one-electron hop

$$R_{ij}(M) = (M \oplus e_i) \oplus e_j.$$

Equivalently,

$$R_{ij}(M) = M - e_i + e_j$$

in occupation-vector notation.

The hop preserves total electronic weight:

$$\mathrm{wt}(R_{ij}M) = \mathrm{wt}(M).$$

Therefore it preserves topological charge:

$$Q_{\mathrm{top}}(Z, R_{ij}M) = Q_{\mathrm{top}}(Z, M).$$

12.2 XOR-diff of a transition

For any two masks M, M', define the transition diff

$$\Delta M = M \oplus M'.$$

Theorem 12.1 (Evenness of electron-number-conserving diffs [Level A])**.** *If*

$$\mathrm{wt}(M) = \mathrm{wt}(M'),$$

then

$$\mathrm{wt}(\Delta M) \equiv 0 \pmod 2.$$

Proof. We have

$$\mathrm{wt}(M \oplus M') = \mathrm{wt}(M) + \mathrm{wt}(M') - 2\,\mathrm{wt}(M \wedge M').$$

If $\mathrm{wt}(M) = \mathrm{wt}(M') = w$, then

$$\mathrm{wt}(M \oplus M') = 2w - 2\,\mathrm{wt}(M \wedge M'),$$

which is even. □

Corollary 12.1 (Minimal radiative patch [Level A])**.** *A nontrivial one-electron hop has*

$$\mathrm{wt}(\Delta M) = 2.$$

Proof. A hop flips exactly one occupied bit to zero and exactly one empty bit to one. □

12.3 Architectural interpretation: radiative and non-radiative diffs

Postulate 12.1 (Spectral diff photon and internal diff transfer [Level B])**.** When an atomic register changes from M to M' while preserving electron number, the transition carries a topological diff

$$\Delta M = M \oplus M'.$$

In a radiative transition this diff is propagated as a photon-like spectral patch with frequency determined by

$$\hbar\omega = E(M) - E(M').$$

In non-radiative transitions, such as Auger decay or internal conversion, the same diff may be transferred virtually to another electronic degree of freedom instead of escaping as a real photon.

Thus, in the IT3 architecture:

Radiative photon = propagating XOR-diff of the electronic mask + spectral phase/frequency.

This gives a finite-register reading of spectral lines:

$$\text{line spectrum} \quad \longleftrightarrow \quad \{\Delta M_\alpha\}$$

where each line corresponds to an allowed layer-hop patch.

Epistemic Guard 12.1. The evenness theorem is [Level A]. The identification of the diff with the emitted photon label is [Level B]. The XOR-diff is a topological label of the transition, not by itself a derivation of the photon wavefunction, polarization, or spectral intensity. The energy functional $E(M)$ remains [Level O].

13 Ionic bonding as closure repair, not literal OR

13.1 Correction of the OR metaphor

A tempting but imprecise statement is:

> Ionic bonding is a bitwise OR of defective pixels.

The exact arithmetic is better described as a *weight-preserving paired data edit* that repairs two prefixes to noble closures.

For sodium and chlorine:

$$Z_{\mathrm{Na}} = 11, \qquad Z_{\mathrm{Cl}} = 17.$$

Neutral audit masks have weights

$$\mathrm{wt}(M_{\mathrm{Na}}) = 11, \qquad \mathrm{wt}(M_{\mathrm{Cl}}) = 17.$$

After one-electron transfer:

$$\mathrm{wt}(M_{\mathrm{Na}^+}) = 10, \qquad \mathrm{wt}(M_{\mathrm{Cl}^-}) = 18.$$

Both are closure numbers:

$$10 = S_1, \qquad 18 = S_2.$$

Total electronic weight is conserved:

$$11 + 17 = 10 + 18 = 28.$$

Topological charges are

$$Q_{\mathrm{top}_{\mathrm{Na}^+}} = 11 - 10 = +1,$$

$$Q_{\mathrm{top}_{\mathrm{Cl}^-}} = 17 - 18 = -1.$$

Hence the pair is neutral:

$$+1 - 1 = 0.$$

Theorem 13.1 (NaCl as exact closure repair [Level A])**.** *The one-electron transfer*

$$(11, 17) \to (10, 18)$$

conserves total electronic weight and maps both fragments to elements of the closure set $\mathcal{S}$*.*

Proof. Direct arithmetic:

$$11 + 17 = 28 = 10 + 18,$$

and

$$10, 18 \in \mathcal{S}.$$

□

13.2 General closure-repair proxy

Define the closure distance

$$\delta(Z) = \min_{s \in \mathcal{S}} |Z - s|.$$

For main-group elements near the $Z = 10$ and $Z = 18$ closures, $\delta(Z)$ reproduces the familiar absolute valence pattern:

Z	Element	$\delta(Z)$	Common valence proxy
3	Li	1	1
4	Be	2	2
5	B	3	3
6	C	4	4
7	N	3	3
8	O	2	2
9	F	1	1
10	Ne	0	0
11	Na	1	1
12	Mg	2	2
13	Al	3	3
14	Si	4	4
15	P	3	3
16	S	2	2
17	Cl	1	1
18	Ar	0	0

Proposition 13.1 (Closure-distance valence proxy [Level A]arithmetic, [Level B]chemical interpretation)**.** *For the displayed main-group range, the integer function $\delta(Z)$ equals the usual absolute valence pattern.*

Proof. Direct minimization against the closure set $\{2, 10, 18\}$ in this range. □

Epistemic Guard 13.1. This is not a derivation of all chemical valences, especially for transition metals, lanthanides, actinides, and relativistic superheavy elements. It is a finite topological proxy.

14 Noble gases as closed-prefix registers

Definition 14.1 (Noble closure [Level A])**.** A nuclear address Z is a noble closure if

$$Z \in \mathcal{S}.$$

The known noble-gas closures in the audit architecture are

$$2,\ 10,\ 18,\ 36,\ 54,\ 86,\ 118.$$

The predicted superheavy closures continue as

$$136,\ 154,\ 162,\ 170,\ 172.$$

For argon:

$$18 = \texttt{0x12} = \texttt{0001 0010}_2.$$

It is the exact end of the second principal closure in the audit sequence.

Proposition 14.1 (Single data edit breaks closure [Level A])**.** *If $s \in \mathcal{S}$, then, away from boundaries,*

$$s - 1 \notin \mathcal{S}, \qquad s + 1 \notin \mathcal{S}.$$

Therefore any one-electron addition or removal from a closed prefix produces a non-closed mask.

Proof. Direct inspection of the finite set $\mathcal{S}$. □

Remark 14.1 (Inertness as closed-overlay stability [Level B]). In the wave-pixel interpretation, a noble gas has no partially filled terminal valence overlay. A one-electron edit creates nonzero Q_{top} and opens a valence channel. Thus inertness is the architectural stability of a closed-prefix pixel.

The phrase "cryptographic protection" may be used as a metaphor, but the strict statement is:

Noble gases are exact closure boundaries of the 172-bit occupation register.

15 Coupling hardware and software: post-transmutation relaxation

The hardware/software split gives a precise statement about what happens immediately after nuclear transmutation.

Suppose a neutral atom has

$$M = M_Z, \qquad Q_{\text{top}}(Z, M) = 0.$$

Apply a hardware address edit

$$H_k : \; Z \to Z - k$$

but keep the electronic mask momentarily unchanged:

$$M' = M.$$

Then

$$Q_{\text{top}}(Z - k, M) = (Z - k) - \text{wt}(M) = Z - k - Z = -k.$$

Thus the daughter system is born with topological charge $-k$.

Theorem 15.1 (Immediate ionization after proton knockout [Level A])**.** *If a neutral atom loses k units of nuclear address while its electronic mask is unchanged, then*

$$Q_{\text{top}} = -k.$$

Neutralization requires a total electronic weight reduction by k:

$$\text{wt}(M) \to \text{wt}(M) - k.$$

Proof. Since initially $\text{wt}(M) = Z$, after $Z \to Z - k$ one has

$$Q_{\text{top}} = (Z - k) - Z = -k.$$

To restore $Q_{\text{top}} = 0$ at address $Z - k$, one needs

$$\text{wt}(M') = Z - k.$$

Thus the electronic weight must decrease by k. □

For lead to gold:

$$Z = 82, \qquad k = 3.$$

Initial neutral state:

$$\text{wt}(M) = 82, \qquad Q_{\text{top}} = 0.$$

Immediately after hardware edit:

$$Z' = 79, \qquad \text{wt}(M) = 82,$$

hence

$$Q_{\text{top}}{}' = 79 - 82 = -3.$$

Final neutral gold state:

$$\mathrm{wt}(M') = 79, \qquad {Q_{\mathrm{top}}}' = 0.$$

The electronic mask diff is

$$M_{82} \oplus M_{79},$$

which in the prefix audit order has Hamming weight

$$\mathrm{popcount}(M_{82} \oplus M_{79}) = 3.$$

Thus:

Nuclear address edit H_3 $\quad\Longrightarrow\quad$ three electronic data deletions for neutralization.

This is the rigorous bridge between the CERN-style transmutation intuition and chemical identity. In the IT3 framework, the nuclear address Z may be read architecturally as a boundary constraint for the discrete electronic standing-wave system. Deallocating three nodes modifies this constraint, forcing the 172-layer electronic system to relax into a new lowest-energy configuration, the Gold mask.

Epistemic Guard 15.1. The Dirichlet-boundary reading is [Level B], not a Level A theorem. The precise discrete Dirac operator, its boundary conditions, and the relaxation functional are [Level O]targets. This paragraph is an architectural interpretation of the exact register edit $M_{82} \to M_{79}$.

16 Spectral-register proofs for selected particle-physics words

The same binary philosophy applies to the particle-physics sector. The Standard-Model observables are not entered as arbitrary decimal parameters; selected ones appear as exact register words. This section makes the bridge more explicit by deriving several register words from the IT3 spectral valency sector.

16.1 Finite algebra and valencies

The IT3 finite algebra contains

$$\mathcal{A}_F = \mathbb{C} \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}).$$

The lepton and quark valencies are

$$v_\ell = 2, \qquad v_q = 3, \qquad v_{\mathrm{total}} = v_\ell + v_q = 5.$$

The bifundamental channel dimension is

$$\dim I = v_\ell v_q = 6.$$

These are Level A integers once the valency assignment is accepted; their physical origin is Level B under the IT3 spectral postulates.

16.2 The $F_{3\text{loop}} = 185$ register word

Theorem 16.1 (Exact three-loop factor [Level A]given valencies)**.**

$$F_{3\text{loop}} = v_{\text{total}}(\dim I)^2 + (v_q^2 - v_\ell^2) = 185.$$

Proof. Direct arithmetic:

$$v_{\text{total}} = 5, \qquad (\dim I)^2 = 6^2 = 36,$$

so

$$v_{\text{total}}(\dim I)^2 = 5 \cdot 36 = 180.$$

Also

$$v_q^2 - v_\ell^2 = 3^2 - 2^2 = 9 - 4 = 5.$$

Therefore

$$F_{3\text{loop}} = 180 + 5 = 185.$$

In hexadecimal,

$$185 = \texttt{0x00B9}.$$

□

16.3 Chern–Simons level and exact $A_{\text{FB}} = 3/13$

The IT3 spectral construction gives the Chern–Simons level

$$k_{\text{CS}} = v_\ell = 2.$$

Define the holographic valency asymmetry parameters

$$\Delta_e = \frac{k_{\text{CS}}}{v_\ell}, \qquad \Delta_q = \frac{k_{\text{CS}}}{v_{\text{total}}}.$$

Then

$$\Delta_e = \frac{2}{2} = 1, \qquad \Delta_q = \frac{2}{5}.$$

For a channel with asymmetry parameter Δ, define

$$A(\Delta) = \frac{4\Delta}{4 + \Delta^2}.$$

Theorem 16.2 (Exact forward-backward asymmetry register [Level A]given the CS level)**.**

$$A_e = A(\Delta_e) = \frac{4}{5}, \qquad A_q = A(\Delta_q) = \frac{5}{13},$$

and therefore

$$A_{\text{FB}} = \frac{3}{4} A_e A_q = \frac{3}{13}.$$

Proof. For electrons:

$$A_e = \frac{4 \cdot 1}{4 + 1^2} = \frac{4}{5}.$$

For quarks:

$$A_q = \frac{4 \cdot (2/5)}{4 + (2/5)^2} = \frac{8/5}{4 + 4/25} = \frac{8/5}{104/25} = \frac{8}{5} \cdot \frac{25}{104} = \frac{200}{520} = \frac{5}{13}.$$

Hence

$$A_{\mathrm{FB}} = \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{13} = \frac{3}{13}.$$

The rational register pair is

$$(3, 13) = \big(\texttt{0x0003}, \texttt{0x000D}\big).$$

□

Epistemic Guard 16.1. The arithmetic is [Level A]once $k_{\mathrm{CS}} = v_\ell = 2$ and the asymmetry formula are accepted. The identification of this register with the physical Z-pole forward-backward asymmetry is [Level B]under the IT3 spectral postulates.

16.4 Fine-structure register

The integer part of the inverse fine-structure constant is

$$137 = \texttt{0x89}.$$

It admits three exact register derivations:

$$137 = 240 - 103,$$

$$137 = 13^2 - 32,$$

$$137 = 136 \vee 1.$$

In hexadecimal:

$$\texttt{0x00F0} - \texttt{0x0067} = \texttt{0x0089},$$

$$\texttt{0x000D}^2 - \texttt{0x0020} = \texttt{0x0089},$$

$$\texttt{0x0088} \vee \texttt{0x0001} = \texttt{0x0089}.$$

Under the IT3 spectral axioms one has the modular expression

$$\Lambda_1 = \sqrt{3}(3 + 2\sqrt{2}), \qquad \Lambda_3 = \phi^2\sqrt{3}, \qquad \phi = \frac{1 + \sqrt{5}}{2},$$

$$N_{\mathrm{twist}} = 103,$$

$$\mathrm{Wr}_{\mathrm{static}} = \Lambda_1^2 - N_{\mathrm{twist}}, \qquad |\mathrm{Wr}|_{\mathrm{dyn}} = |\mathrm{Wr}_{\mathrm{static}}| \left(1 + \frac{1}{5N_{\mathrm{twist}}}\right).$$

The geometric gauge coupling register is

$$\alpha_{\mathrm{geom}}^{-1} = \Lambda_1^2 + \frac{N_{\mathrm{twist}}}{v_q} + \frac{\Lambda_3}{v_\ell v_q} + \frac{|\mathrm{Wr}|_{\mathrm{dyn}}}{v_\ell v_q v_{\mathrm{total}}}.$$

Numerically,

$$\alpha_{\mathrm{geom}}^{-1} \approx 137.037.$$

The truncated integer part is the exact Level A register word

$$\lfloor \alpha_{\mathrm{geom}}^{-1} \rfloor = 137 = \texttt{0x89}.$$

Epistemic Guard 16.2. The integer identities are [Level A]. The full spectral formula for $\alpha_{\mathrm{geom}}^{-1}$ is [Level B]under the IT3 spectral axioms. The statement that this register is the physical inverse fine-structure constant is not proved here from the wave-pixel architecture alone.

16.5 Z' mass word

The IT3 spectral sector gives the heavy-vector resonance structure

$$M_{Z'} = \eta(D'_\gamma)\, M_{\mathrm{bulk}}\, E_{\mathrm{egg}}^{-1}\, \det{}_F(I - \alpha_{\mathrm{geom}} P_\gamma).$$

In the rank-one projection sector,

$$\det{}_F(I - \alpha_{\mathrm{geom}} P_\gamma) = 1 - \alpha_{\mathrm{geom}}.$$

The boundary linking number gives

$$\eta(D'_\gamma) = v_\ell = 2.$$

The egg-shell energy factor is

$$E_{\mathrm{egg}} = \sqrt{3} - \sqrt{2}, \qquad E_{\mathrm{egg}}^{-1} = \sqrt{3} + \sqrt{2},$$

because

$$(\sqrt{3} - \sqrt{2})(\sqrt{3} + \sqrt{2}) = 3 - 2 = 1.$$

Therefore

$$M_{Z'} = 2\, M_{\mathrm{bulk}}\, (\sqrt{3} + \sqrt{2})\, (1 - \alpha_{\mathrm{geom}}).$$

With the IT3 spectral value of M_{bulk}, this gives the truncated mass-bus word

$$M_{Z'} \approx 1088 \text{ GeV} = \texttt{0x0440}.$$

Epistemic Guard 16.3. The algebraic identity $E_{\mathrm{egg}} E_{\mathrm{egg}}^{-1} = 1$ is [Level A]. The spectral formula for $M_{Z'}$ is [Level B]under the IT3 spectral axioms. The numerical value 1088 GeV is a truncated/rounded register word, not an experimental claim.

16.6 Baryon residual register

For the doubly charmed baryon, the IT3 spectral baryon calculation gives a linear estimate

$$M_{\Xi_{cc}}^{\mathrm{lin}} \approx 3619 \text{ MeV} = \texttt{0x0E23},$$

while the experimental value is approximately

$$M_{\Xi_{cc}}^{\mathrm{exp}} \approx 3621 \text{ MeV} = \texttt{0x0E25}.$$

The truncated integer residual is

$$3621 - 3619 = 2 = \texttt{0x0002}.$$

Epistemic Guard 16.4. The hexadecimal identities are [Level A]. The interpretation of the residual as a higher-order dynamical correction is [Level B].

16.7 Confinement ratio

The strict confinement bound is

$$\frac{S_{\mathrm{macro}}}{S_{\mathrm{micro}}} = \frac{72}{24} = 3.$$

In hexadecimal:

$$\frac{\texttt{0x0048}}{\texttt{0x0018}} = \texttt{0x0003}.$$

Architecturally,

$$3 = v_q.$$

Epistemic Guard 16.5. The integer ratio is [Level A]. The identification with the quark valency and confinement bound is [Level B]under the IT3 spectral postulates.

17 Unified register table

In the table below, $w = \mathrm{wt}(M)$ denotes the electronic occupation weight.

Object	Hex / binary / formula	IT3 meaning	Status
Vacuum mask	`0xE100`	$57600 = 240^2$, four active bits	[Level A]
Dormant mask	`0x1EFF`	twelve dormant payload bits	[Level A]
Kummer wall	`0x000D`	13, illegal 3-bit squareclass opcode	[Level A]
Photon edge	`0x0057`	central rising edge	[Level A]
Antiphoton edge	`0x0056`	central falling/domain edge	[Level A]
Source sum	`0x00AD`	$86 + 87 = 173 = 172 + 1$	[Level A]
FCC closure	`0x00AC`	$172 = 43 \ll 2$	[Level A]
Ogg boundary	`0x002B`	43, first Heegner prime outside Monster set	[Level A]
Alpha register	`0x0089`	$137 = 240 - 103$	[Level A]integer part
A_{FB} numerator	`0x0003`	3	[Level A]
A_{FB} denominator	`0x000D`	Kummer wall 13	[Level A]
CS level	$k_{\mathrm{CS}} = 2$	lepton valency v_ℓ	[Level B]
Δ_e	1	boundary electron asymmetry parameter	[Level A]/[Level B]
Δ_q	2/5	bulk quark asymmetry parameter	[Level A]/[Level B]
$F_{3\mathrm{loop}}$	`0x00B9`	$185 = 5 \cdot 36 + 5$	[Level A]given valencies
$\alpha_{\mathrm{geom}}^{-1}$ spectral	≈ 137.037	modular gauge-coupling register	[Level B]
Z' mass word	`0x0440`	1088 GeV truncated resonance code	[Level B]/[Level C]
Ξ_{cc} linear	`0x0E23`	3619 MeV code	[Level B]
Ξ_{cc} experiment	`0x0E25`	3621 MeV code	[Level B]
Ξ_{cc} residual	`0x0002`	truncated higher-order shift	[Level B]
Confinement ratio	`0x0003`	$72/24 = 3 = v_q$	[Level A]/[Level B]
Pb address	`0x0052`	$Z = 82$	[Level A]
Au address	`0x004F`	$Z = 79$	[Level A]
Pb->Au address friction	`0x001D`	popcount(29) = 4	[Level A]
Topological charge	$Z - w$	address-data mismatch	[Level A]/[Level B]
Neutron edit	$N \to N - m$	volume/ECC padding	[Level B]/[Level C]
Strain proxy	$Z^2/(Z+N)$	active-node density tension	[Level A]/[Level B]
Asymmetry proxy	$(N-Z)/(Z+N)$	neutron padding buffer	[Level A]/[Level B]
Chiral projector	$(1-\epsilon_\ell)/2$	one-oriented lepton strand	[Level A]/[Level B]
Wilson loop	$\mathrm{Tr}\,\mathcal{P}e^{i\oint A}$	global torus holonomy	[Level B]/[Level O]
Electron capture	$\Delta Z = -1, \Delta w = -1$	synchronized HW/SW edit	[Level A]
Beta-minus	$\Delta Z = +1, \Delta w = 0$	residual cation	[Level A]
UPC neutron knockout	$\Delta Z = 0, \Delta w = 0$	chemical identity fixed	[Level A]
Auger	$\Delta Z = 0, \Delta w = -2$	double software deletion	[Level A]
Ionization diff	e_i	one-bit software edit	[Level A]
Radiative hop diff	$e_i \oplus e_j$	two-bit spectral patch	[Level A]
Na closure repair	$11 \to 10$	cation formation	[Level A]
Cl closure repair	$17 \to 18$	anion formation	[Level A]

18 Machine-checkable audit code

The following Python code verifies the load-bearing identities of the wave-pixel, the three-register state space, synchronized hardware/software edits, neutron padding and strain proxies, the Pb $\to$ Au binary edit, NaCl closure repair, spectral diff parity, noble closures, chiral valency counting, chiral projector parity flip, Chern–Simons rational register, torus mode sanity, and selected particle-physics registers.

Listing 1: IT3-WavePixel v1.0 audit

```
#!/usr/bin/env python3
from fractions import Fraction

def popcount(x: int) -> int:
    return bin(x).count("1")

```

```
def hex16(x: int) -> str:
    return f"0x{x & 0xFFFF:04X}"

def bin8(x: int) -> str:
    return format(x & 0xFF, "08b")

def bin16(x: int) -> str:
    return format(x & 0xFFFF, "016b")

def mask_int(Z: int) -> int:
    """
    Prefix occupation mask: first Z bits are 1.
    Returns an integer whose binary representation has Z low bits set.
    """
    if not (0 <= Z <= 172):
        raise ValueError("Z must be between 0 and 172")
    return (1 << Z) - 1

def Q_top(Z: int, M: int) -> int:
    """
    Topological charge: nuclear address minus electronic occupation
       weight.
    """
    return Z - popcount(M)

def Q_weight(Z: int, w: int) -> int:
    """
    Topological charge from address and electronic weight.
    """
    return Z - w

def delta_Q(dZ: int, dw: int) -> int:
    """
    Change in residual topological charge.
    """
    return dZ - dw

def strain(Z: int, N: int) -> Fraction:
    """
    Active-node strain proxy: Z^2 / (Z+N).
    """
    if Z + N == 0:
        raise ValueError("Z+N must be positive")
    return Fraction(Z * Z, Z + N)

def asymmetry(Z: int, N: int) -> Fraction:
    """
    Neutron padding asymmetry: (N-Z) / (Z+N).
    """
    if Z + N == 0:
        raise ValueError("Z+N must be positive")
    return Fraction(N - Z, Z + N)

def Pi_active(eps: int) -> Fraction:
    """
    Chiral lepton strand projector: (1 - eps) / 2.
    eps = -1 -> active strand projector = 1.
    eps = +1 -> active strand projector = 0.
```

```
    """
    if eps not in (-1, 1):
        raise ValueError("eps must be +/-1")
    return Fraction(1 - eps, 2)

def mode_norm2(mode) -> Fraction:
    """
    Discrete torus norm:
    n1^2 + n2^2/2 + n3^2/3
    = (6 n1^2 + 3 n2^2 + 2 n3^2) / 6.
    """
    n1, n2, n3 = mode
    return Fraction(6 * n1 * n1 + 3 * n2 * n2 + 2 * n3 * n3, 6)

def cycle_coeff(mode, cycle) -> int:
    """
    Integer holonomy coefficient for a general torus cycle:
    n1*m1 + n2*m2 + n3*m3.
    """
    return sum(n * m for n, m in zip(mode, cycle))

def cs_asym(D: Fraction) -> Fraction:
    """
    Chern-Simons asymmetry factor:
    A(D) = 4D / (4 + D^2).
    """
    return Fraction(4) * D / (Fraction(4) + D * D)

# ------------------------------------------------------------
# 1. 172-layer wave pixel capacities
# ------------------------------------------------------------
def kappa(n):
    return min(n, 11 - n)

def q(n):
    return (kappa(n) + 3) // 2

C = [2 * q(n) * q(n) for n in range(12)]

assert C == [2, 8, 8, 18, 18, 32, 32, 18, 18, 8, 8, 2]
assert sum(C) == 172
assert sum(C[:6]) == 86
assert 172 == 43 << 2

# Cumulative closures
S = []
acc = 0
for c in C:
    acc += c
    S.append(acc)

assert S == [2, 10, 18, 36, 54, 86, 118, 136, 154, 162, 170, 172]

S_set = set(S)

# ------------------------------------------------------------
# 2. Central binary edge
# ------------------------------------------------------------
```

```
A_edge = 86  # antiphoton / classical equator domain
P_edge = 87  # photon / central edge domain

assert A_edge == 0x56
assert P_edge == 0x57
assert A_edge ^ P_edge == 1
assert A_edge + P_edge == 173
assert A_edge + P_edge == 0xAD
assert 173 == 172 + 1

# no hexadecimal nibble carry in 0x56 + 0x57
assert ((P_edge & 0xF) + (A_edge & 0xF)) < 16
assert (((P_edge >> 4) & 0xF) + ((A_edge >> 4) & 0xF)) < 16

# ------------------------------------------------------------
# 3. Pb -> Au hardware address edit
# ------------------------------------------------------------
Z_PB = 82
Z_AU = 79
DELTA_P = 3

assert Z_PB - DELTA_P == Z_AU

assert Z_PB == 0x52
assert Z_AU == 0x4F

assert Z_PB - DELTA_P == 0x4F

XOR_PB_AU_ADDRESS = Z_PB ^ Z_AU
TFI_PB_AU_ADDRESS = popcount(XOR_PB_AU_ADDRESS)

assert XOR_PB_AU_ADDRESS == 0x1D
assert XOR_PB_AU_ADDRESS == 29
assert TFI_PB_AU_ADDRESS == 4

# Floor location: both in floor 5, because S[4]=54 and S[5]=86
def floor_of(Z):
    for n, Sn in enumerate(S):
        if Z <= Sn:
            return n
    raise ValueError("Z out of range")

assert floor_of(Z_PB) == 5
assert floor_of(Z_AU) == 5

offset_PB = Z_PB - S[4]
offset_AU = Z_AU - S[4]

assert offset_PB == 28
assert offset_AU == 25

# Stable gold from Pb-208 requires 3 protons and 8 neutrons removed
assert 208 - 3 - 8 == 197
assert 82 - 3 == 79
assert 126 - 8 == 118

# ------------------------------------------------------------
# 4. Hardware edit followed by software relaxation
```

```
# ------------------------------------------------------------
M_Pb_neutral = mask_int(82)
M_Au_neutral = mask_int(79)

assert popcount(M_Pb_neutral) == 82
assert popcount(M_Au_neutral) == 79

assert Q_top(82, M_Pb_neutral) == 0

# Immediately after nuclear address edit, electrons unchanged:
assert Q_top(79, M_Pb_neutral) == -3

# Final neutral gold:
assert Q_top(79, M_Au_neutral) == 0

# Electronic mask diff has weight 3:
assert popcount(M_Pb_neutral ^ M_Au_neutral) == 3

# ------------------------------------------------------------
# 5. Synchronized hardware/software edits
# ------------------------------------------------------------
# Generic starting neutral register:
Z0 = 82
w0 = 82
assert Q_weight(Z0, w0) == 0

# Electron capture: Z -> Z-1, w -> w-1
assert delta_Q(-1, -1) == 0
assert Q_weight(Z0 - 1, w0 - 1) == 0

# Beta-minus: Z -> Z+1, bound w unchanged
assert delta_Q(+1, 0) == +1
assert Q_weight(Z0 + 1, w0) == +1

# Beta-plus: Z -> Z-1, bound w unchanged initially
assert delta_Q(-1, 0) == -1
assert Q_weight(Z0 - 1, w0) == -1

# Alpha residual: Z -> Z-2, bound w unchanged initially
assert delta_Q(-2, 0) == -2
assert Q_weight(Z0 - 2, w0) == -2

# Proton knockout residual: Z -> Z-3, bound w unchanged initially
assert delta_Q(-3, 0) == -3
assert Q_weight(Z0 - 3, w0) == -3

# Photoionization: Z unchanged, w -> w-1
assert delta_Q(0, -1) == +1
assert Q_weight(Z0, w0 - 1) == +1

# Electron attachment: Z unchanged, w -> w+1
assert delta_Q(0, +1) == -1
assert Q_weight(Z0, w0 + 1) == -1

# Auger double ionization: Z unchanged, w -> w-2
assert delta_Q(0, -2) == +2
assert Q_weight(Z0, w0 - 2) == +2

```

```
# ------------------------------------------------------------
# 6. Neutron ECC padding, strain and asymmetry proxies
# ------------------------------------------------------------
N0 = 126
A_mass0 = Z0 + N0
assert A_mass0 == 208

# Neutron knockout: N -> N-1, Z and w unchanged
assert Q_weight(Z0, w0) == 0
assert Z0 + (N0 - 1) == 207

# Charge-density proxy chi = Z / (Z+N)
chi_Pb208 = Fraction(Z0, Z0 + N0)
chi_unpadded = Fraction(Z0, Z0 + Z0)

assert chi_Pb208 < chi_unpadded
assert chi_Pb208 == Fraction(82, 208)
assert chi_unpadded == Fraction(1, 2)

# Strain proxy decreases when neutron padding increases.
assert strain(82, 126) < strain(82, 100) < strain(82, 82)

# Asymmetry proxy increases when neutron padding increases.
assert asymmetry(82, 126) > asymmetry(82, 100) > asymmetry(82, 82)

# Sample stable-ish nuclides: N/Z should be nondecreasing in this toy
    audit set.
samples = [
    (2, 2),      # He-4
    (6, 6),      # C-12
    (26, 30),    # Fe-56
    (82, 126),   # Pb-208
]

ratios = [Fraction(N, Z) for Z, N in samples]
assert ratios == sorted(ratios)

# Explicit values:
assert ratios[0] == 1
assert ratios[1] == 1
assert ratios[2] == Fraction(30, 26)
assert ratios[3] == Fraction(126, 82)

# ------------------------------------------------------------
# 7. Topological charge examples with masks
# ------------------------------------------------------------
# Neutral Na:
M_Na = mask_int(11)
assert Q_top(11, M_Na) == 0

# Na+ by removing one electronic layer:
M_Na_plus = mask_int(10)
assert Q_top(11, M_Na_plus) == 1
assert popcount(M_Na ^ M_Na_plus) == 1

# Neutral Cl:
M_Cl = mask_int(17)
assert Q_top(17, M_Cl) == 0
```

```python

# Cl- by adding one electronic layer:
M_Cl_minus = mask_int(18)
assert Q_top(17, M_Cl_minus) == -1
assert popcount(M_Cl ^ M_Cl_minus) == 1

# NaCl closure repair:
assert 11 + 17 == 10 + 18
assert 10 in S_set
assert 18 in S_set
assert Q_top(11, M_Na_plus) + Q_top(17, M_Cl_minus) == 0

# ------------------------------------------------------------
# 8. Spectral diff parity
# ------------------------------------------------------------
# One-electron hop preserves weight and gives XOR diff of weight 2.
def hop(M: int, i: int, j: int) -> int:
    """
    Move one occupation from bit i to bit j.
    Assumes bit i is occupied and bit j is empty.
    Uses XOR so no bitwise NOT is needed.
    """
    if ((M >> i) & 1) != 1:
        raise ValueError("source bit not occupied")
    if ((M >> j) & 1) != 0:
        raise ValueError("target bit not empty")
    return M ^ (1 << i) ^ (1 << j)

M_test = mask_int(20)
for i in range(20):
    for j in range(20, 40):
        M2 = hop(M_test, i, j)
        assert popcount(M2) == 20
        assert popcount(M_test ^ M2) == 2

# General theorem check for several same-weight masks:
def same_weight_xor_even(M1, M2):
    assert popcount(M1) == popcount(M2)
    return popcount(M1 ^ M2) % 2 == 0

assert same_weight_xor_even(mask_int(5), mask_int(5))
assert same_weight_xor_even(hop(mask_int(10), 0, 20), mask_int(10))

# ------------------------------------------------------------
# 9. Noble closures and closure-distance valence proxy
# ------------------------------------------------------------
noble_known = [2, 10, 18, 36, 54, 86, 118]

for s in noble_known:
    assert s in S_set
    if s - 1 >= 0:
        assert (s - 1) not in S_set
    if s + 1 <= 172:
        assert (s + 1) not in S_set

def closure_distance(Z: int) -> int:
    return min(abs(Z - s) for s in S)

```

```
expected_valence_proxy = {
    3: 1,   # Li
    4: 2,   # Be
    5: 3,   # B
    6: 4,   # C
    7: 3,   # N
    8: 2,   # O
    9: 1,   # F
    10: 0,  # Ne
    11: 1,  # Na
    12: 2,  # Mg
    13: 3,  # Al
    14: 4,  # Si
    15: 3,  # P
    16: 2,  # S
    17: 1,  # Cl
    18: 0,  # Ar
}

for z, v in expected_valence_proxy.items():
    assert closure_distance(z) == v

# Argon closure in binary:
assert 18 == 0x12
assert bin(18) == "0b10010"

# ------------------------------------------------------------
# 10. Kummer wall and squareclass register
# ------------------------------------------------------------
WALL13 = 13
ALLOWED_SF = {1, 2, 3, 5, 6, 10, 15, 30}

assert WALL13 == 0x0D
assert WALL13 > 0b111
assert WALL13 not in ALLOWED_SF

# ------------------------------------------------------------
# 11. Chiral valency counting, projector parity flip,
#     F3-loop, and Chern-Simons rational register
# ------------------------------------------------------------
v_l = 2
v_q = 3
v_total = v_l + v_q
dim_I = v_l * v_q

assert v_l == 2
assert v_q == 3
assert v_total == 5
assert dim_I == 6

F3loop = v_total * dim_I * dim_I + (v_q * v_q - v_l * v_l)
assert F3loop == 185
assert F3loop == 0x00B9

# Chiral projector:
# eps = -1 selects active strand, eps = +1 rejects it.
assert Pi_active(-1) == 1
assert Pi_active(1) == 0
```

```
assert Pi_active(-1) != Pi_active(1)

# The weak charged current is architecturally routed through one oriented
# strand of the two-strand lepton bundle. This is a Level B reading, not
# a derivation of the chosen orientation.
active_lepton_strands = 1
assert active_lepton_strands < v_l

# Chern-Simons level:
k_CS = v_l
assert k_CS == 2

Delta_e = Fraction(k_CS, v_l)
Delta_q = Fraction(k_CS, v_total)

assert Delta_e == 1
assert Delta_q == Fraction(2, 5)

A_e = cs_asym(Delta_e)
A_q = cs_asym(Delta_q)

assert A_e == Fraction(4, 5)
assert A_q == Fraction(5, 13)

A_FB_cs = Fraction(3, 4) * A_e * A_q
assert A_FB_cs == Fraction(3, 13)

# ------------------------------------------------------------
# 12. Torus mode label and Wilson-holonomy sanity check
# ------------------------------------------------------------
# A global torus mode is labelled by an integer triple.
mode = (2, 1, 3)

# Squared norm in the discrete torus spectrum:
# n1^2 + n2^2 / 2 + n3^2 / 3
assert mode_norm2(mode) == Fraction(15, 2)

# General single-valuedness:
# k dot lambda = 2*pi*(n1*m1 + n2*m2 + n3*m3).
# For the diagonal cycle (1,1,1), the coefficient is n1+n2+n3.
assert cycle_coeff(mode, (1, 1, 1)) == 6

# For a non-diagonal cycle, the coefficient is the integer pairing.
assert cycle_coeff(mode, (1, 2, 3)) == 13

# Wilson-loop holonomy is a global function of the loop class.
# Here we only audit that the mode label is globally well-defined.
loop_class = mode
assert loop_class == (2, 1, 3)

# ------------------------------------------------------------
# 13. Fine-structure register
# ------------------------------------------------------------
ALPHA_INT = 137
KISS240 = 240
NTWIST = 103
```

```
CAP32 = 32
SIGMA136 = 136

assert ALPHA_INT == 0x89
assert ALPHA_INT == KISS240 - NTWIST
assert ALPHA_INT == WALL13 * WALL13 - CAP32
assert ALPHA_INT == SIGMA136 | 1
assert popcount(ALPHA_INT) == 3

# ------------------------------------------------------------
# 14. A_FB rational register
# ------------------------------------------------------------
AFB = Fraction(3, 13)
assert AFB.numerator == 3
assert AFB.denominator == 13
assert abs(float(AFB) - 0.23077) < 1e-5
assert AFB == A_FB_cs

# ------------------------------------------------------------
# 15. Particle-physics mass/residual registers
# ------------------------------------------------------------
ZPRIME = 1088
assert ZPRIME == 0x0440

XI_LIN = 3619
XI_EXP = 3621
XI_RESID = XI_EXP - XI_LIN

assert XI_LIN == 0x0E23
assert XI_EXP == 0x0E25
assert XI_RESID == 0x0002

SMACRO = 72
SMICRO = 24
CONF3 = SMACRO // SMICRO

assert SMACRO == 0x0048
assert SMICRO == 0x0018
assert CONF3 == 3 == 0x0003
assert CONF3 == v_q

# ------------------------------------------------------------
# 16. Output
# ------------------------------------------------------------
print("=" * 78)
print(" IT3-WAVEPIXEL v1.0 THREE-REGISTER + STRAIN + CHIRAL + HOLONOMY
    AUDIT ")
print("=" * 78)

print("172-layer capacities:")
print(C)
print("cumulative closures:")
print(S)
print()

print("Central edge:")
print(f"86 = 0x56 = {bin8(86)}")
print(f"87 = 0x57 = {bin8(87)}")
```

```
print(f"86 XOR 87 = {hex16(86 ^ 87)}")
print(f"86 + 87 = {hex16(86 + 87)} = 173 = 172 + 1")
print()

print("Pb -> Au hardware address edit:")
print(f"Pb Z=82: {hex16(82)} {bin8(82)}")
print(f"Au Z=79: {hex16(79)} {bin8(79)}")
print(f"0x52 - 0x03 = {hex16(0x52 - 0x03)}")
print(f"0x52 XOR 0x4F = {hex16(XOR_PB_AU_ADDRESS)}")
print(f"TFI_address(Pb->Au, delta Z=-3) = {TFI_PB_AU_ADDRESS}")
print(f"floor(Pb) = {floor_of(82)}, offset = {offset_PB}")
print(f"floor(Au) = {floor_of(79)}, offset = {offset_AU}")
print(f"stable Au path: 208 - 3p - 8n = {208 - 3 - 8}")
print()

print("Hardware/software relaxation:")
print(f"Q_top(82, M_82) = {Q_top(82, M_Pb_neutral)}")
print(f"Q_top(79, M_82) = {Q_top(79, M_Pb_neutral)}")
print(f"Q_top(79, M_79) = {Q_top(79, M_Au_neutral)}")
print(f"electronic mask diff weight = {popcount(M_Pb_neutral ^
    M_Au_neutral)}")
print()

print("Synchronized edits:")
print(f"electron capture delta Q = {delta_Q(-1, -1)}")
print(f"beta-minus delta Q = {delta_Q(+1, 0)}")
print(f"beta-plus delta Q = {delta_Q(-1, 0)}")
print(f"alpha residual delta Q = {delta_Q(-2, 0)}")
print(f"UPC neutron knockout delta Q = {delta_Q(0, 0)}")
print(f"Auger double ionization delta Q = {delta_Q(0, -2)}")
print()

print("Neutron ECC padding and strain proxies:")
print(f"Pb-208 chi = Z/A = {chi_Pb208} = {float(chi_Pb208):.6f}")
print(f"unpadded N=Z chi = {chi_unpadded} = {float(chi_unpadded):.6f}")
print(f"strain(82,126) = {strain(82,126)} = {float(strain(82,126)):.3f}
    ")
print(f"strain(82,82)  = {strain(82,82)} = {float(strain(82,82)):.3f}")
print(f"asymmetry(82,126) = {asymmetry(82,126)} = {float(asymmetry
    (82,126)):.6f}")
print("sample N/Z ratios:", [float(r) for r in ratios])
print()

print("NaCl closure repair:")
print(f"Na neutral weight = {popcount(M_Na)}")
print(f"Na+ weight = {popcount(M_Na_plus)}, Q_top = {Q_top(11,
    M_Na_plus)}")
print(f"Cl neutral weight = {popcount(M_Cl)}")
print(f"Cl- weight = {popcount(M_Cl_minus)}, Q_top = {Q_top(17,
    M_Cl_minus)}")
print(f"total weight before = {popcount(M_Na) + popcount(M_Cl)}")
print(f"total weight after  = {popcount(M_Na_plus) + popcount(
    M_Cl_minus)}")
print()

print("Spectral diff parity:")
print("one-electron hop XOR weight = 2 for tested hops")
print()
```

```
576
577 print("Chiral valency counting, F3-loop, and projector:")
578 print(f"v_l = {v_l}, v_q = {v_q}, dim(I) = {dim_I}")
579 print(f"F3loop = {F3loop} = {hex16(F3loop)}")
580 print(f"Pi_active(-1) = {Pi_active(-1)}")
581 print(f"Pi_active(+1) = {Pi_active(1)}")
582 print("parity flip changes projector: verified")
583 print()
584
585 print("Chern-Simons rational register:")
586 print(f"k_CS = {k_CS}")
587 print(f"Delta_e = {Delta_e}")
588 print(f"Delta_q = {Delta_q}")
589 print(f"A_e = {A_e}")
590 print(f"A_q = {A_q}")
591 print(f"A_FB = {A_FB_cs} = {float(A_FB_cs):.6f}")
592 print()
593
594 print("Torus mode / Wilson-holonomy sanity:")
595 print(f"mode = {mode}")
596 print(f"mode_norm2 = {mode_norm2(mode)} = {float(mode_norm2(mode)):.6f}
    ")
597 print(f"diagonal-cycle coefficient = {cycle_coeff(mode, (1,1,1))}")
598 print(f"general cycle (1,2,3) coefficient = {cycle_coeff(mode, (1,2,3))
    }")
599 print()
600
601 print("Particle-physics registers:")
602 print(f"alpha_inv_int = {hex16(137)} = 137 = 240 - 103")
603 print(f"A_FB = {AFB.numerator}/{AFB.denominator} = {float(AFB):.6f}")
604 print(f"Zprime mass word = {hex16(1088)} = 1088 GeV")
605 print(f"Xi residual = {hex16(2)} = 2 MeV truncated code")
606 print(f"Confinement = {hex16(72)} / {hex16(24)} = {hex16(3)}")
607 print()
608
609 print("STATUS: Level A identities passed. Level B/C physical
    interpretation conditional on IT3 postulates.")
610 print("=" * 78)
```

19 Falsifiable bets and experimental tests

19.1 Bet 1: absolute pixel ceiling

The IT3 wave-pixel predicts an absolute closure at:

$$Z = 172.$$

The next source node is

$$173 = \texttt{0xAD} = 172 + 1,$$

whose low nibble is

$$\texttt{0xD} = 13,$$

the first illegal squareclass opcode:

$$\sqrt{13} \notin \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}).$$

Refutation condition: if a neutral bound atom with $Z \geq 173$ exhibits ordinary periodic structure, the 172-node FCC mask topology is falsified.

19.2 Bet 2: transmutation as address edit

For any reaction changing nuclear charge by $\Delta Z = k$, the IT3 audit assigns a binary address friction

$$\mathrm{TFI}_k(Z) = \mathrm{popcount}(Z \oplus (Z - k)).$$

For

$$\mathrm{Pb} \to \mathrm{Au} + 3p,$$

one has

$$\mathrm{TFI}_3(82) = 4.$$

This does not by itself predict a cross-section. It predicts that the event belongs to the same binary borrow-cascade family as α-decay and proton radioactivity.

19.3 Bet 3: post-transmutation electronic relaxation

If a neutral atom loses k protons and the electronic mask is initially unchanged, the immediate topological charge is

$$Q_{\mathrm{top}} = -k.$$

Neutralization requires removal of k electronic occupation units.
For lead to gold:

$$k = 3, \qquad Q_{\mathrm{top}} = -3, \qquad \Delta\,\mathrm{wt}(M) = -3.$$

Refutation condition: if a clean isolated three-proton knockout from a neutral lead-like system systematically failed to produce the expected three-unit electronic relaxation signature, the hardware/software coupling would be challenged.

19.4 Bet 4: neutron knockout does not change chemical identity

For any pure neutron edit

$$V_m : \; N \to N - m,$$

with Z and M unchanged,

$$\Delta Q_{\mathrm{top}} = 0.$$

Thus UPC neutron knockout, isotopic substitution, and neutron capture/change should not alter the elemental address.

Refutation condition: if a controlled pure neutron-number change systematically produced a different neutral chemical element without proton-number change, the three-register architecture would fail.

19.5 Bet 5: neutron strain-padding trend

The strain proxy

$$\Sigma_{\text{strain}}(Z, N) = \frac{Z^2}{Z + N}$$

decreases when neutron padding increases at fixed Z. The asymmetry proxy

$$\eta_{\text{asym}}(Z, N) = \frac{N - Z}{Z + N}$$

increases with neutron padding.

The IT3 reading predicts that heavier stable nuclei should require progressively larger N/Z.

Refutation condition: if a substantial population of heavy stable nuclei with $Z \gg 20$ and $N \leq Z$ were discovered, the simple strain-padding architecture would be falsified.

19.6 Bet 6: spectral diff parity

Any electron-number-conserving radiative transition should have an even XOR-diff in the occupation mask:

$$\text{wt}(M_{\text{before}} \oplus M_{\text{after}}) \equiv 0 \pmod 2.$$

A one-electron hop has minimal diff weight two.

This is an architectural prediction about the representation of transitions, not yet a direct spectral-line prediction.

19.7 Bet 7: closure-distance valence proxy

For main-group elements near the closures 10 and 18, the integer function

$$\delta(Z) = \min_{s \in \mathcal{S}} |Z - s|$$

reproduces the common valence pattern

$$1, 2, 3, 4, 3, 2, 1, 0.$$

Refutation condition: if a chemically well-characterized main-group element systematically violated this closure-distance pattern without a recognized relativistic or correlation explanation, the simple proxy would fail.

19.8 Bet 8: chiral projector parity violation

If the weak charged current is routed by

$$\Pi_\ell^{\text{act}} = \frac{1 - \epsilon_\ell}{2},$$

then parity reversal $\epsilon_\ell \mapsto -\epsilon_\ell$ changes the projector. Therefore the charged weak sector cannot be parity invariant.

Refutation condition: if a fundamental charged-current interaction coupling equally to both lepton-helix orientations were discovered, the chiral hardware-thread model would fail.

19.9 Bet 9: Wilson-holonomy entanglement and no-signalling

If entanglement is a shared global torus holonomy, then correlations may be nonlocal in the Bell sense, but local pointer edits must not permit controllable faster-than-light signalling.

Refutation condition: if reproducible faster-than-light communication were experimentally demonstrated, any global-thread/no-signalling interpretation, including the IT3 Wilson-holonomy reading, would be falsified.

19.10 Bet 10: Chern–Simons rational register

If the IT3 Chern–Simons level is

$$k_{\mathrm{CS}} = v_\ell = 2,$$

then the exact rational register prediction is

$$A_{\mathrm{FB}} = \frac{3}{13}.$$

Refutation condition: if high-precision Z-pole data excluded 3/13 beyond the stated dynamical corrections and known systematics, the CS/valency register interpretation would fail.

19.11 Bet 11: particle-physics register words

The register audit gives concrete numerical targets:

$$\alpha_{\mathrm{geom}}^{-1} \text{ integer part} = 137 = \texttt{0x89},$$

$$A_{\mathrm{FB}} = 3/13 \approx 0.23077,$$

$$M_{Z'} \approx 1088 \text{ GeV} = \texttt{0x0440},$$

$$M_{\Xi_{cc}}^{\mathrm{lin}} \approx 3619 \text{ MeV}, \qquad M_{\Xi_{cc}}^{\mathrm{exp}} \approx 3621 \text{ MeV}.$$

Refutation condition: if future high-precision data exclude these rational/integer register structures beyond the stated dynamical corrections, the corresponding IT3 register interpretation fails.

20 Conclusion

The three-register extension

$$(Z, N, M)$$

makes the IT3-WavePixel architecture substantially stronger.

Nuclear transmutation is a hardware operation:

$$H_k : \ Z \to Z - k.$$

Neutron changes are volume/ECC operations:

$$V_m : \ N \to N - m, \qquad \Delta Q_{\mathrm{top}} = 0.$$

Ionization, excitation, and bonding are software operations:

$$D_i : \ M \to M \oplus e_i, \qquad R_{ij} : \ e_i \to e_j.$$

The topological charge

$$Q_{\text{top}} = Z - \text{wt}(M)$$

is the exact mismatch between allocated nuclear address and occupied electronic layers. The lead-to-gold channel

$$\text{Pb } (Z = 82) \to \text{Au } (Z = 79) + 3p$$

is the binary address edit

$$\texttt{0x52} \to \texttt{0x4F},$$

with address friction

$$\text{popcount}(82 \oplus 79) = \text{popcount}(29) = 4.$$

If the electronic mask is momentarily unchanged, the daughter has

$$Q_{\text{top}} = -3,$$

and neutralization requires three electronic data deletions:

$$\text{wt}(M) : 82 \to 79.$$

For stable ^{197}Au from ^{208}Pb, the neutron padding must also change:

$$208 - 3p - 8n = 197.$$

The neutron is best read not as an address bit but as a volume/ECC padding coordinate. This explains why neutron knockout does not change chemistry, while proton knockout does. The strain proxy

$$\Sigma_{\text{strain}} = \frac{Z^2}{Z + N}$$

gives a finite-register diagnostic for why heavy nuclei need more neutrons per proton.

Spectral transitions preserve electron number and therefore have even XOR-diffs. A one-electron hop has minimal diff weight two. The photon is architecturally the propagating spectral patch carrying this diff, while its frequency remains a dynamical quantity to be derived from the spectral operator. Non-radiative transitions, such as Auger decay, carry the same topological diff internally rather than emitting a real photon.

Ionic bonding is not literally a bitwise OR. It is a weight-preserving paired data edit that repairs defective prefixes to noble closures:

$$(11, 17) \to (10, 18).$$

Noble gases are exact closed-prefix states:

$$Z \in \mathcal{S}.$$

The chiral lepton thread $v_\ell = 2$ gives an algebraic projector

$$\Pi_\ell^{\text{act}} = \frac{1 - \epsilon_\ell}{2}$$

whose parity flip is exact. This gives a register-level language for parity violation, but the dynamical origin of the selected orientation remains open.

Torus holonomy and Wilson loops give a global-register language for entanglement. The correct statement is not that entanglement disappears, but that it is re-described as nonseparability of a shared global mode/holonomy, compatible with no-signalling.

The spectral-register bridge is strengthened by exact valency arithmetic:

$$F_{3\text{loop}} = 185 = \texttt{0x00B9},$$

$$k_{\text{CS}} = 2, \qquad \Delta_e = 1, \qquad \Delta_q = \frac{2}{5},$$

$$A_e = \frac{4}{5}, \qquad A_q = \frac{5}{13}, \qquad A_{\text{FB}} = \frac{3}{13},$$

$$137 = \texttt{0x89} = \texttt{0xF0} - \texttt{0x67},$$

$$M_{Z'} \leftrightarrow 1088 = \texttt{0x0440},$$

$$\Xi_{cc} \text{ residual} = 2 = \texttt{0x0002},$$

$$S_{\text{macro}}/S_{\text{micro}} = 72/24 = 3 = \texttt{0x0003}.$$

The honest final statement is:

IT3-WavePixel v1.0 is a finite binary architecture in which nuclear physics edits the address, neutrons edit th

The next dynamical step is to derive the canonical order of the 172 modes, the energy functional $E(M)$, the neutron strain threshold, the weak chiral projector, and the global entanglement structure from a spectral action. Until then, the arithmetic audit is exact, while the physical interpretation remains explicitly conditional.

A Compact register trace sheet

Listing 2: Compact trace sheet

```
# IT3-WavePixel v1.0 trace sheet

# 172-layer pixel
C = [2,8,8,18,18,32,32,18,18,8,8,2]
assert sum(C) == 172
assert sum(C[:6]) == 86
assert 172 == 43 << 2

# closures
S = [2,10,18,36,54,86,118,136,154,162,170,172]
assert 10 in S
assert 18 in S

# central edge
assert 86 == 0x56
assert 87 == 0x57
```

```
assert 86 ^ 87 == 1
assert 86 + 87 == 173 == 0xAD

# Pb -> Au hardware address edit
assert 82 == 0x52
assert 79 == 0x4F
assert 0x52 - 0x03 == 0x4F
assert 82 ^ 79 == 29 == 0x1D
assert bin(29).count("1") == 4

# stable gold requires neutron padding edit
assert 208 - 3 - 8 == 197

# hardware/software relaxation
def mask_int(Z): return (1 << Z) - 1
def popcount(x): return bin(x).count("1")
def Q_top(Z, M): return Z - popcount(M)
def Q_weight(Z, w): return Z - w
def delta_Q(dZ, dw): return dZ - dw

assert Q_top(82, mask_int(82)) == 0
assert Q_top(79, mask_int(82)) == -3
assert Q_top(79, mask_int(79)) == 0
assert popcount(mask_int(82) ^ mask_int(79)) == 3

# synchronized edits
assert delta_Q(-1, -1) == 0  # electron capture
assert delta_Q(+1, 0) == +1  # beta-minus residual
assert delta_Q(-1, 0) == -1  # beta-plus residual
assert delta_Q(-2, 0) == -2  # alpha residual
assert delta_Q(0, 0) == 0    # neutron knockout
assert delta_Q(0, -2) == +2  # Auger double ionization

# neutron strain proxies
from fractions import Fraction
def strain(Z, N): return Fraction(Z * Z, Z + N)
def asymmetry(Z, N): return Fraction(N - Z, Z + N)

assert strain(82, 126) < strain(82, 82)
assert asymmetry(82, 126) > asymmetry(82, 82)

# NaCl closure repair
assert 11 + 17 == 10 + 18
assert Q_top(11, mask_int(10)) == 1
assert Q_top(17, mask_int(18)) == -1
assert popcount(mask_int(11) ^ mask_int(10)) == 1
assert popcount(mask_int(18) ^ mask_int(17)) == 1

# spectral diff parity
M = mask_int(20)
M2 = M ^ (1 << 0) ^ (1 << 30)
assert popcount(M) == popcount(M2)
assert popcount(M ^ M2) == 2

# chiral valency counting, F3-loop, and projector
v_l, v_q = 2, 3
v_total = v_l + v_q
dim_I = v_l * v_q
```

```
assert v_total == 5
assert dim_I == 6
assert v_total * dim_I * dim_I + (v_q * v_q - v_l * v_l) == 185

def Pi_active(eps): return Fraction(1 - eps, 2)
assert Pi_active(-1) == 1
assert Pi_active(1) == 0
assert Pi_active(-1) != Pi_active(1)

# Chern-Simons rational register
k_CS = v_l
Delta_e = Fraction(k_CS, v_l)
Delta_q = Fraction(k_CS, v_total)
def cs_asym(D): return Fraction(4) * D / (Fraction(4) + D * D)
A_e = cs_asym(Delta_e)
A_q = cs_asym(Delta_q)
A_FB = Fraction(3, 4) * A_e * A_q
assert Delta_e == 1
assert Delta_q == Fraction(2, 5)
assert A_e == Fraction(4, 5)
assert A_q == Fraction(5, 13)
assert A_FB == Fraction(3, 13)

# torus mode sanity
def mode_norm2(mode):
    n1, n2, n3 = mode
    return Fraction(6 * n1 * n1 + 3 * n2 * n2 + 2 * n3 * n3, 6)

def cycle_coeff(mode, cycle):
    return sum(n * m for n, m in zip(mode, cycle))

assert mode_norm2((2, 1, 3)) == Fraction(15, 2)
assert cycle_coeff((2, 1, 3), (1, 1, 1)) == 6
assert cycle_coeff((2, 1, 3), (1, 2, 3)) == 13

# Kummer wall
assert 13 == 0x0D
assert 13 > 0b111

# fine structure register
assert 137 == 0x89
assert 137 == 240 - 103
assert 137 == 13**2 - 32
assert 137 == 136 | 1

# particle registers
assert 1088 == 0x0440
assert 3621 - 3619 == 2
assert 185 == 0x00B9
assert 72 // 24 == 3
```

B Possible extensions

1. Full table of all 172 elemental masks in a physically motivated subshell order.
2. Rendering of the 172-layer pixel as a 12-row binary image.

3. Comparison of $\mathrm{TFI}_k(Z)$ with measured partial cross sections for proton, neutron, alpha, and fragment emission.
4. Derivation of the canonical order from a discrete Dirac operator on $T^3/\mathbb{Z}_2$.
5. Bridge to NIST atomic spectra: spectral lines as layer-hop operators $P_m H P_n$.
6. Explicit computation of ΔM supports for hydrogen-like transitions in the audit ordering.
7. Extension of closure-distance valence proxy to d-block and f-block with subclosure sets.
8. Derivation of neutron magic numbers from a discrete ECC/lattice-resonance model.
9. Rigorous spectral derivation of the chiral weak projector from the orientation of the $v_\ell = 2$ lepton bundle.
10. Wilson-loop computation of entanglement phases on noncontractible T^3 cycles.
11. Quantum-information interpretation: the 172-bit pixel as a holographic error-correcting code, with entanglement as shared global syndrome.
12. Full dynamical derivation of $E(M)$, the neutron rupture threshold, and the real-photon spectral density from the IT3 spectral action.

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